

# Why a Steepest-Descent Line Search at $k = 0$ is Consistent with *Minimization of expensive functions in the presence of forbidden regions*

## 1 The degeneracy at $k = 0$

`ffjm2` minimizes  $f(x) = \frac{1}{2} \|F(x)\|^2$ , where  $F = (f_1, \dots, f_m)^\top$  is the residual vector. At external iteration  $k$  it keeps one symmetric matrix  $H_i^k$  per residual component  $f_i$ , approximating  $\nabla^2 f_i(x^k)$ , and solves

$$M_k(s) = \sigma \|s\|^2 + \frac{1}{2} \sum_i q_i(s)^2, \quad q_i(s) = f_i(x^k) + \nabla f_i(x^k)^\top s + \frac{1}{2} s^\top H_i^k s, \quad (1)$$

accepting or rejecting  $x^k + s$  by the ratio between actual and predicted reduction, and growing/shrinking  $\sigma$  accordingly (this is the regularization scheme of Algorithm 4.1 of [1], implemented in `ffjm2` with an empirical persistent- $\sigma$  variant — the paper resets  $\sigma \leftarrow 0$  at the start of every iteration and never shrinks it, but persisting it between iterations and shrinking it on acceptance performed better in practice).

At  $k = 0$ , every  $H_i^0 = 0$  (no secant pair has been observed yet), so (1) collapses to

$$M_0(s) = \sigma \|s\|^2 + \frac{1}{2} \sum_i (f_i(x^0) + \nabla f_i(x^0)^\top s)^2, \quad (2)$$

a single damped Gauss–Newton subproblem. Two things are worth noting about this particular instance of the subproblem:

1. Unlike every subsequent iteration,  $\sigma = \sigma_0$  here is a fixed constant, not yet informed by any accept/reject test on *this* problem. Whether it makes (2) behave like Gauss–Newton ( $\sigma \rightarrow 0$ ) or like scaled steepest descent ( $\sigma \rightarrow \infty$ , where the minimizer of (2) is asymptotically parallel to  $-\nabla f(x^0)$ ) is essentially a guess about the local curvature scale of  $f$  at  $x^0$  that the method has had no chance to calibrate yet.
2. There is, by construction, no curvature information at all to justify solving a quartic (rather than a quadratic) model here: the quartic term  $\frac{1}{2} s^\top H_i^k s$  is identically zero for every  $i$ .

So the very first step is the one iteration where the method’s own machinery (quasi-Newton curvature, calibrated regularization) has nothing to work with.

## 2 A self-contained argument: the closed form of $M_0$

Rather than appealing to a different algorithm, the cleanest justification comes directly out of (2) itself, which is an unconstrained, strictly convex quadratic in  $s$  and can be solved in closed form. Write  $J := F'(x^0)$  for the Jacobian of  $F$  at  $x^0$  (the matrix whose  $i$ -th row is  $\nabla f_i(x^0)^\top$ ), so that  $g := \nabla f(x^0) = J^\top F(x^0)$ , and  $A := J^\top J$  (symmetric positive semidefinite). Then

$$\nabla_s M_0(s) = (2\sigma I + A) s + g = 0 \implies s(\sigma) = -(A + 2\sigma I)^{-1} g.$$

**Lemma A (descent for every  $\sigma > 0$ ).** Since  $A + 2\sigma I \succ 0$  for  $\sigma > 0$ , its inverse is symmetric positive definite, hence

$$g^\top s(\sigma) = -g^\top (A + 2\sigma I)^{-1} g < 0 \quad \text{whenever } g \neq 0.$$

So  $s(\sigma)$  is a genuine descent direction for  $f$  for every  $\sigma > 0$ , not just asymptotically — the  $\sigma$ -loop never has a “wrong-direction” problem at  $k = 0$ ; what it can have is a badly-sized step, which is what the accept/reject test and the growth of  $\sigma$  are actually correcting for.

**Proposition B (alignment with  $-g$  as  $\sigma \rightarrow \infty$ ).** Factor  $2\sigma$  out of the inverse:

$$s(\sigma) = -(A + 2\sigma I)^{-1}g = -\frac{1}{2\sigma} \left( I + \frac{A}{2\sigma} \right)^{-1} g.$$

As  $\sigma \rightarrow \infty$ ,  $A/(2\sigma) \rightarrow 0$ , and matrix inversion is continuous at the identity, so  $(I + A/(2\sigma))^{-1} \rightarrow I$ . Hence

$$s(\sigma) = -\frac{1}{2\sigma} g + o\left(\frac{1}{\sigma}\right) \implies \frac{s(\sigma)}{\|s(\sigma)\|} \longrightarrow -\frac{g}{\|g\|} \quad \text{as } \sigma \rightarrow \infty. \quad (3)$$

In words: for  $\sigma$  large enough, the solution of the degenerate  $k = 0$  subproblem (2) already is, to leading order, a steepest-descent step of length  $\approx 1/2\sigma$  — the two are asymptotically the same computation. This is a direct consequence of (2)’s own closed form, entirely internal to Algorithm 4.1.

In practice, the rejection ladder  $\sigma_0, \sigma_{j+1} = \gamma\sigma_j$  ( $\gamma > 1$  the growth factor) tries  $s(\sigma_0), s(\sigma_1), \dots$  until one is accepted. By Proposition B, once  $\sigma_j \gg \|A\|/2$  the successive trials are already essentially parallel to  $-g$  and shrink by the fixed factor  $1/\gamma$  — so once it reaches that scale, the ladder is already behaving like a geometric backtracking line search along  $-g$ . Below that scale, though — which includes  $\sigma_0$  itself whenever  $\sigma_0$  (a fixed constant, unrelated to the problem at hand) is not already  $\gg \|A\|/2$  —  $s(\sigma_0)$  is a perfectly valid descent direction (Lemma A) but generally *not* aligned with  $-g$ , and its length is not calibrated to  $\|g\|$  at all: whether it is accepted on the first try is essentially a coincidence of scale, not a controlled property of the method.

Replacing the  $k = 0$  subproblem by an explicit Armijo line search along  $-g$  therefore does not change what the method is doing once  $\sigma$  is large — it is, to leading order, the same computation. What it removes is the uncalibrated, possibly wasted, low- $\sigma$  rejections at the start of the ladder, by searching directly along the direction (3) shows the ladder is heading toward anyway, with a step-length search (quadratic interpolation) that is more sample-efficient than a fixed geometric factor  $1/\gamma$ . It is also computationally cheaper: each rejected  $\sigma$  in the ladder re-invokes the general-purpose numerical solver used for the full quartic subproblem (which explores several random restarts to guard against spurious local minima), even though at  $H_i = 0$  the subproblem has the trivial closed form  $s(\sigma) = -(A + 2\sigma I)^{-1}g$  above — a backtracking line search only needs cheap residual evaluations  $F(x^0 + \alpha p)$  to test sufficient decrease.

## References

- [1] F. A. Fortunato and J. M. Martínez, *Minimization of expensive functions in the presence of forbidden regions*, unpublished manuscript, 2026. ([\\_research/ffjm2.pdf](#) in this repository.)